X360 x LLBox

WhatsApp Chatbot Release Documentation

Version 2.0: response to the LLBox x X360 full test report of 4-5 October 2026. What changed, how the bot now behaves for drivers and box owners, how every new API is used, and the test evidence.

Prepared for	LLBox (product and QA)
Prepared by	X360
Date	7 October 2026
Baseline	LLBox x X360 WhatsApp bot full test report (4-5 Oct 2026): driver items R1-R4, owner findings F1-F19, Part F developer guide (driver-contact, peek, COD, box health, box number, delivery status, report-issue, preferences, support contact, per-integration limits)
System	WhatsApp Cloud API -> n8n workflow 'LLBox SUB - Operations' -> X360 backend (conversation-turn) -> LLBox Agent API v1 (production)
Status	R1-R4 fixed (4/4). F1-F19 fixed (19/19). All Part F APIs integrated. 121 of 121 automated checks pass. Live on production since 7 October 2026.

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1. Summary

LLBox's report said the driver side was mostly fixed and the owner side was the main gap, and asked us to integrate the new APIs for cash on delivery, dead boxes and oversized parcels, always sending conversation_id. All of it is done and running on the production number:

- Driver side. Roman Hindi problems are now recognised (R1), box photos arrive in order (R2), spaced references are read whole (R3), and '???', emoji, stickers and Urdu get a proper reply (R4).
- New driver cases. COD, dead box and oversized parcel follow LLBox's F11 flows: peek first where useful, then driver-contact with conversation_id, and the customer's number is read back once to that driver only.
- Owner mode (F1-F19). A recognised number now gets a real owner assistant: menu, register (several references, dates, which box, COD), list, status, cancel with confirmation, guest and replacement codes, box status, box number, reports and tickets, STOP/START, driver-call consent, support.
- After the handover. The bot polls delivery status and thanks the driver when the keypad completes the delivery.

Test suite	Checks	How it runs	Result
Original 30 driver scenarios (30 Sep report)	43 / 43	Live LLBox lookups; notify-recipient and driver-contact stubbed	**PASS**
Driver edge cases: R1, R3, R4, COD, dead box, oversized, privacy	33 / 33	LLBox stubbed (no codes consumed, no numbers released); real model	**PASS**
Owner mode F1-F19 on the QA owner account	33 / 33	Live identify, deliveries, register, cancel, box-status, support; codes, tickets, preferences stubbed	**PASS**
Two-box owner (F8) and R4 wording	12 / 12	identify and owner endpoints stubbed	**PASS**
Total	**121 / 121**		**PASS**

Live checks on production in addition: a driver handover sent through the real WhatsApp webhook (text first, then four photos in order), and a real driver-contact call with an unknown reference (404 not_available, confirming conversation_id is accepted).

2. Status of every report item

2.1 Driver items still open after the previous round (report A4)

#	Finding	What the bot does now	Evidence	Status
R1	Problems written in Roman Hindi / Hinglish were ignored (handover given, customer never notified).	Rules now cover Roman Hindi problem phrases (code nahi chal raha, box khul nahi raha, jagah nahi hai, bahut bada hai, box nahi mil raha, phone nahi utha raha, light nahi jal rahi, dabba kharab ho gaya ...). Anything the rules miss is classified by a small model (gpt-4o-mini, temperature 0, 6 s timeout) into one of LLBox's issue types. The model only classifies; it never writes the reply.	driver edge R1a-R1l (14 checks)	**PASS**
R2	Box photos arrived in random order.	The n8n 'Send WhatsApp Box Photos' node now sends one photo at a time with 2.5 s between sends, in API array order, after the text reply. Verified live on WhatsApp: street, building, front door, box.	live run, execution 10487	**PASS**
R3	'spl 88 123456' was looked up as '123456'.	Adjacent groups are joined into one reference when a short prefix is followed by digit groups (at least 5 digits, not phone-like). A word before a complete reference is never glued on ('raha SPL77100205' stays 'SPL77100205').	driver edge R3a-R3c, cleanRef checks; multi-box R4	**PASS**
R4	Impersonation wording; spaced form told 'another driver has the code'; Urdu policy; '???' Arabic-only; emoji gave an almost empty bubble.	Spaced and compact forms are now the same reference, so the same driver is never told another driver has the code. Owner-claim wording now asks the owner to write from the number linked to their account. '???', emoji and stickers get a full reply in both languages when no language is known; reactions get no reply. Urdu script gets the English and Arabic reply.	driver edge E1-E3; multi-box R4 (3 checks); 30-scenario #28a-d	**PASS**

2.2 Owner findings (report B3)

#	Prio	Finding	What the bot does now	Evidence	Status
F1	P0	'cancel X' registered X and said Done.	Intent is decided before references. A message with a cancel verb never registers. Cancel = look up by reference (deliveries with reference filter), confirm, then deliveries/cancel with delivery_id.	owner: F1 x4	**PASS**
F2	P0	Previous reference reused when the new one was invalid or missing.	Conversation memory (slot) has a 15-minute expiry, is cleared after every completed action and is only used to answer the bot's own last question. A bare date after a finished action asks for the order number.	owner: F2 stale ref	**PASS**
F3	P0	Owner reporting a stuck driver was handled as a driver.	Any number identify recognises gets owner mode. 'The driver says the code doesn't work for X' issues a replacement code with delivery_id. Only 'I am the driver' switches a recognised number to driver mode.	owner: F15, F4	**PASS**
F4	P0	identify called on 5 of ~90 messages.	identify is called live on every message. The last good answer is used only if the live call fails.	all owner checks	**PASS**
F5	P1	List requests deflected to the app.	List uses whatsapp/deliveries: matched first, then by day, 10 shown with a 'more' count, optional this/next week or a single day; offers to cancel.	owner: F7 x2	**PASS**
F6	P1	Cancel requests deflected to support.	Cancel by reference (one or several), or pick from the cancellable list; always confirm; clear wording for already cancelled, too late (driver has code), finished, not found.	owner: F6 x2, F1	**PASS**

#	Prio	Finding	What the bot does now	Evidence	Status
F7	P1	Code and status got the driver template; no battery/online.	Guest code (whatsapp/code without delivery), replacement code (with delivery_id), box status (box-status: parcel inside, expected today vs comfortable/most per day, online/last seen, battery, how full).	owner: F13-F15	**PASS**
F8	P1	Multi-box never asked.	With two or more boxes the bot asks which box (number, nickname, code, or home/office words) for register, guest code and reports, unless the message already names one.	multi-box: 9 checks	**PASS**
F9	P1	'What number do I give Noon?' could not be answered.	Answers with the box's phone_number from identify; when null, with the bot's WhatsApp number (LLBOX_DRIVER_WHATSAPP_NUMBER, +1 507 710 7553), plus how couriers use it.	owner: F9	**PASS**
F10	P2	Relative and past dates ignored; unknown day defaulted silently.	Date parser: today/tomorrow/day after, weekdays (this/next), in N days/weeks, next week/month, weekend, dd/mm, '12 October', ISO, Arabic forms. Past dates are refused and the day asked again; the day is asked once if missing; 'not sure' is accepted.	owner: F10 x2	**PASS**
F11	P2	More than 120 days reported as 'may be in the past'.	Says the date is more than 120 days away and asks again.	owner: F11	**PASS**
F12	P2	Three references in one message: only the first handled.	Up to 5 references per message, one line each; a full day is reported once for all affected references and all of them are kept for the next answer.	owner: F12, F10 relative	**PASS**
F13	P2	'yes', 'menu', '2', 'what can you do' not understood.	Numbered menu built from available_actions; digits, yes/no and 'menu/help/what can you do' work in English and Arabic.	owner: F7 digit, F5	**PASS**
F14	P2	Rambling question answered with 'use the app'.	Intent router reads the whole message; unknown messages get 'didn't quite get that' plus the menu, never a deflection.	owner: F5	**PASS**
F15	P2	Add member, change phone, delete account, opt-out, theft had no route.	Opt-out/in and driver-contact consent go to preferences. Account changes and theft become report-issue tickets (category other / theft) with the ticket number read back.	owner: F17 x3, F18, F19	**PASS**
F16	P2	'Red light, box not opening' answered as a code question.	Becomes a box_fault ticket (report-issue); duplicates within 10 minutes are not re-filed.	owner: F16 x2	**PASS**
F17	P2	'Sunday' on a Sunday registered for today without confirming.	Registers for today and adds: 'That's today. If you meant next Sunday, send "next Sunday".' 'next Sunday' means next week.	date parser unit check	**PASS**
F18	P2	Owner told to visit llbox.ai.	Removed. Owners get the owner product answer; no website push.	owner: F3 greeting	**PASS**
F19	P2	English warning printed an ISO date.	All days are written as 'Thursday 8 October' / 'الخميس 8 أكتوبر' (Riyadh time); the busy-day warning is localised.	owner: F2, F12	**PASS**

Evidence names refer to the checks listed in Appendix B. F9 and F15/F16 were API gaps in the report; they are closed with LLBox's new phone_number, preferences and report-issue endpoints.

3. How the bot works now

3.1 Path of one message

1. WhatsApp -> n8n. Meta webhook into workflow 'LLBox SUB - Operations' (WIULGDZMvW2e7p25). Reactions and delivery receipts are dropped; images and voice notes are read; stickers pass through as input type 'sticker'.
2. n8n -> backend. POST /api/tools/conversation-turn with phone, text, input type, image reading and message id.
3. Duplicate guard. The same text from the same number within 5 seconds gets one reply.
4. Read the message. Normalise digits and spacing, extract references (spaced forms joined, dimensions and phone numbers excluded), detect language, classify driver problems and owner intents.
5. identify. Live on every message. Recognised account -> owner mode, unless the message says 'I am the driver'.
6. Owner or driver flow. Fixed, reviewed wording only; facts come only from the LLBox API answer for this message.
7. Reply. One language when the sender's language is known, both (Arabic first for Arabic speakers) otherwise. n8n sends the text, then the box photos one at a time.
8. Follow-ups. A background worker in the backend polls delivery status and contact reviews and sends the result to the driver on WhatsApp.

3.2 Who is talking: owner or driver

- identify runs on every message (report F4/F5). If user_found is true the message goes to owner mode, with owner wording only. If identify fails, the last good answer for that number (up to 30 minutes old) is used; with none, the message is treated as a driver message, which never reveals account data.
- A recognised owner who writes 'I am the driver ...' is handled as a driver for that message (owners who deliver themselves). 'My driver ...' stays in owner mode.
- Everyone else is a driver. A driver who claims to be the owner is asked to write from the number linked to the LLBox account; nothing about the box is disclosed.

3.3 Language

- English or Arabic replies mirror the sender. Arabizi and Arabic script get Arabic first, then English.
- When there is nothing to detect ('???', emoji, a sticker) and no earlier language in the last 24 hours, the reply is in both languages. Urdu script gets English and Arabic.
- Days are always written as words ('Thursday 8 October', 'الخميس 8 أكتوبر') in Riyadh time.

3.4 Conversation memory

- Owner memory (slot): what the bot last asked (day, box, COD, pick, confirm, description, menu) with the references involved. Expires after 15 minutes; cleared when the action completes; a new intent always wins over an old slot.
- Driver memory: the problem waiting for an order number (10 minutes), the parcel-size question, the last handovers (3 days, for 'where is the box again' and two-number messages), and one retry marker per reference (20 minutes) so 'still not working' after 'try once more' escalates instead of looping.
- Neither ever holds a box code or a customer phone number.

4. Driver flows

4.1 Normal handover

1. Driver sends an order number (typed, spaced, or a label photo).
2. Driver lookup (by-reference) returns box, address, map and a one-time code; the bot sends the handover text, then the box photos one by one.
3. If LLBox marks the order cash on delivery, the handover adds: 'This order is marked cash on delivery. If you need to collect payment first, tell me and I will connect you with the customer.'
4. A delivery-status follow-up is scheduled; when the keypad completes the delivery the driver gets a thank-you.

4.2 Problems at the box

Driver says	API calls	Bot does
Code not working / box dead / no light	driver-contact box_unresponsive	200: customer first name + number; 202: 'try once more, I'm checking' then follow-up; 409 box_online: 'try all six digits and wait for the beep', a second report notifies the customer; other: customer notified.
Parcel too big	peek, then driver-contact parcel_too_large	Size known: ask L x W x H in cm, then call with parcel_cm. 200: number; 409 parcel_fits: 'turn it onto its smallest side', second report notifies; 422: ask size.
Cash on delivery	peek, then driver-contact cod	Prepaid: 'no cash to collect, use the box'. 200: number and amount. 403: customer prefers no calls, has been told. 404: ask for the full order number. 429: message passed to the customer instead.
Box full	notify-recipient box_full	'Don't force the lid', customer notified.
Can't find the box	lookup + notify cannot_find_box	Address, map and photos again, customer notified.
Customer not answering / wants the number	notify-recipient other	'I can't share the number, but I've let the customer know'.
Leave with guard / neighbour	notify-recipient other	'Deliveries are completed only at the LLBox', customer notified.
Problem without an order number	-	Asks for the order number, remembers the problem for 10 minutes.

4.3 Roman Hindi, spaced references, stickers

- Rules cover the common Roman Hindi forms of every problem type. When no rule matches and the message looks like a problem (a negation or problem word, plus Roman Hindi or at least 8 words), the model classifies it. The model's answer only selects which fixed flow runs.
- Spaced references: 'spl 88 123456', 'NQ 4455 6677 88' are one reference; 'flat 12 554433' gives 554433; dimensions ('60 x 40 x 30') and phone numbers are never references.
- Stickers: 'I can't read stickers, send the order number or a photo of the label'. Emoji only: delivery emoji -> ask for the order number; thanks emoji after a handover -> thanks. WhatsApp reactions get no reply.

5. Owner mode

5.1 What the owner can say

Owner says (EN or AR)	Bot does	LLBox API
Hi / hello / ابحرم	Greeting with name, short summary and the numbered menu	identify
1 / register / a bare order number	Ask the day once, which box if several, register, then ask COD	register (expected-deliveries)
2 / what's coming / where is X	List, or the status of the given references	whatsapp/deliveries
3 / cancel X / I don't need X	Look up, confirm, cancel	deliveries + deliveries/cancel
4 / code for my cleaner / guest	One-time guest code for the box	whatsapp/code
The driver's code doesn't work for X	Replacement code linked to the delivery	deliveries + whatsapp/code (delivery_id)
5 / is my box ok / anything inside	Parcel inside, expected today, online, battery, full %	box-status
6 / what number do I give Noon	Box number (or the bot's number) and how couriers use it	identify
7 / red light / stolen / damaged	Ticket with reference read back; theft advice	report-issue
Delete my account / change my phone / add member	Passed to the LLBox team as a ticket	report-issue (other)
STOP / START	Opt out / back in	preferences
Don't let drivers call me / they can call me	Driver-contact consent off / on	preferences
Support / talk to a person	Support details if configured, otherwise the app hint	support-contact
I am the driver ...	Switches this message to the driver flow	-

5.2 Rules that protect the owner

- A message with a cancel word never registers anything. Every cancel is confirmed first ('Cancel X (expected Friday 16 October)? Reply yes'); 'no' keeps it.
- A reference is never reused silently: memory only answers the bot's own last question and expires after 15 minutes.
- The day is asked once when missing; 'not sure' is accepted and LLBox's default applies. Past dates and dates more than 120 days ahead are refused with the reason. 'Sunday' on a Sunday is today, and the reply points out that 'next Sunday' means next week.
- When a day is full, the reply gives the limit and asks for another day for all affected references at once. A busy (but not full) day gets the 'comfortably holds 3, at most 5' warning.
- Registration asks once about cash on delivery; 'COD 200 for all of them' applies to every reference just registered.
- Status of an overdue delivery (expected day passed, still pending) offers to cancel it.
- With several boxes the bot asks which one, unless the message names it (number, nickname, box code, home/office).
- Report tickets are not duplicated within 10 minutes; theft adds advice to contact the store or courier.

6. API integration (report Part F)

Base URL https://llbox.up.railway.app/api/v1, header X-Agent-API-Key. Every call that changes something (driver-contact, cancel, code, report-issue) is never retried automatically, so a slow answer cannot create a second ticket or a second code.

Part F / endpoint	Used when	How each answer is handled
F1 POST /agent/driver-contact	Driver with COD, dead box or oversized parcel, after an order number	Always sends conversation_id = 'x360-' + 24 hex of SHA-256 of the driver's number (stable, not reversible). 200 name + number read back once; 202 retry + poll; 403 opted out; 404 ask for full number; 409 box_online / parcel_fits retry advice then notify on second try, prepaid / not_cod -> no cash, ambiguous -> normal lookup; 422 ask for size; 429 pass a message instead, never loop.
GET /agent/driver-contact/:id	After 202 pending_review	Polled every 30 s for 30 minutes by the follow-up worker; released -> number sent once; refused -> told; timeout -> customer notified instead.
F2 GET /agent/box/by-reference/:ref?peek=true	Before COD / oversized contact	Prepaid -> no contact needed; max_parcel_cm known -> ask the parcel size first; never consumes a code.
F3 payment_type / cod_amount	Owner registration; driver handover	Asked once after registering ('COD with amount, prepaid, or not sure'); 'not sure' leaves it unknown; re-sent to update. Driver handover adds a COD note when the order is cod.
F4 box_health, max_parcel_cm	Owner box status; dead box; oversized	Online/offline/out of service with last seen and battery; full %; size used to decide whether to ask dimensions.
F5 identify (phone_number, health, preferences)	Every message	Owner mode, box number, multi-box question, greeting summary.
F6 POST /agent/whatsapp/deliveries (filters)	List, status, cancel, replacement code	reference, date_from/date_to, status[], limit.
POST /agent/whatsapp/deliveries/cancel	Owner confirmed a cancel	200, 409 too late / finished, 404.
POST /agent/whatsapp/code	Guest or replacement code	Code shown once, never re-read; 422 asks which box.
POST /agent/whatsapp/box-status	'Is my box ok / anything inside'	See F4.
F7 GET /agent/delivery/status/:ref	After every handover; on 'done/delivered'	Polled from 3 minutes after handover (every 3 min, then 10 min, up to 6 h); thanks the driver once completed.
F8 POST /agent/whatsapp/report-issue	Box fault, theft, damaged parcel, account requests	Reads back the ticket id; 10-minute duplicate guard; 422 asks which box.
F9 POST /agent/whatsapp/preferences	STOP / START / drivers may (not) call me	Strict booleans.
GET /agent/support-contact	Support / human requests	Cached 10 minutes. configured=false -> says no support line is published yet and gives the app hint.
F10 Per-integration limits	All calls	429 is never retried in a loop; the driver is offered a message instead.

7. Privacy and safety rules

- The customer's phone number is only obtained through driver-contact, only after the driver gave an order number, only shown to that driver in that conversation, and never stored or logged. A test checks no stored conversation contains it.
- The bot no longer reads recipient_whatsapp_number from the driver lookup, so LLBox can switch DRIVER_LOOKUP_INCLUDE_PHONE off. 44 customer numbers that earlier lookups had saved in our database were deleted.
- Box codes appear only in the reply that issues them; they are never re-read, logged in clear, or stored in conversation memory.
- The bot never says whether a number has an LLBox, never gives an address without a valid order number, and never invents facts such as ETAs or support numbers.
- conversation_id is a one-way hash of the driver's number, so LLBox's guessing lock works per driver without receiving the number itself.
- Opt-out is honoured: owners are never messaged unprompted by the bot; follow-ups go only to drivers who just wrote to us.

8. Follow-up worker

LLBox has no webhooks, so the backend runs a small worker (every 20 seconds) over follow-up jobs stored in MongoDB. Jobs are locked while processed, so two backend processes cannot send twice.

Job	Created	Polls	Ends with
Delivery status	After every driver handover	delivery/status from 3 minutes after the handover, every 3 minutes, every 10 minutes after 20 tries, for 6 hours	completed: thank-you to the driver; cancelled / expired / not found: stop; 429: retry in 5 minutes
Contact review	driver-contact answered 202 pending_review	driver-contact/:id every 30 seconds for 30 minutes	released: number sent once; opted out / refused: told; timeout: customer notified instead

Follow-up messages go only to drivers who wrote to us in the last few hours, so they stay inside WhatsApp's 24-hour window. Jobs never contain a phone number or a code.

9. Configuration and deployment

9.1 Backend settings (backend/.env; values not shown)

Setting	Purpose
LLBOX_BASE_URL, LLBOX_API_KEY	LLBox Agent API base URL and X-Agent-API-Key
LLBOX_DRIVER_WHATSAPP_NUMBER	Number given to owners for couriers when the box has no phone_number (new)
OPENAI_API_KEY, ISSUE_CLASSIFIER_MODEL	Problem classifier for Roman Hindi and long messages (default gpt-4o-mini)
WHATSAPP_TOKEN, WHATSAPP_PHONE_NUMBER_ID	Follow-up messages sent by the backend (Graph API v21.0)
FOLLOWUPS_DISABLED	true stops the follow-up worker (used by the test suites)
MONGODB_URI	Conversation memory (TurnState), follow-up jobs, ops records
LLBOX_DUMMY_MODE	Must be false in production

9.2 Running and deploying

- Backend: PM2 process `llbox-backend` on 127.0.0.1:5100. Restart with `pm2 restart llbox-backend --update-env`; health at `GET /health`; logs in `backend/logs/combined.log` ([turn], [owner], [contact], [followup], [classify] lines).

- n8n: `python3 n8n/fix_operations_workflow.py` patches and stores a new workflow version; then `n8n publish:workflow` with that version id and restart the n8n container. Every version is kept in workflow_history for rollback.
- Current production: backend restarted 7 October 2026 with all changes; n8n version f575fd8c (sequential photos, reaction skip).

10. Testing

Suite (backend/scripts)	What it covers	Live vs stubbed	Result
turn-regression.js	Original 30 driver scenarios (30 Sep report)	Live LLBox lookups; notify-recipient and driver-contact stubbed	43/43
driver-edge-regression.js	Driver edge cases: R1, R3, R4, COD, dead box, oversized, privacy	LLBox stubbed (no codes consumed, no numbers released); real model	33/33
owner-regression.js	Owner mode F1-F19 on the QA owner account	Live identify, deliveries, register, cancel, box-status, support; codes, tickets, preferences stubbed	33/33
multibox-regression.js	Two-box owner (F8) and R4 wording	identify and owner endpoints stubbed	12/12

Run any suite with `FOLLOWUPS_DISABLED=true node scripts/<suite>.js --json=out.json`. The owner suite takes the owner number as an argument and cancels every test delivery it creates. No suite sends WhatsApp messages. The driver suites never consume real codes for edge cases and never release real numbers.

What is tested live and what is not

- Live: identify, deliveries list/filters, register with dates and COD, cancel, box-status, support-contact, driver lookup and notify for the original 30 scenarios, one real WhatsApp handover with photos, one real driver-contact shape check.
- Stubbed: driver-contact outcomes (200/202/403/404/409/422/429), guest/replacement codes, report-issue and preferences in the owner suite, and the two-box owner. These would otherwise notify LLBox's tester, change the tester's account or release a real number.

11. Open items and notes for LLBox

- Box number (LLBox to confirm). QA-PROD-TEST-BOX has phone_number null, so owners are told +1 507 710 7553 (the bot's WhatsApp number). Please confirm this is the number customers should give couriers, or fill phone_number per box.
- Support line. support-contact returns configured=false; the bot says so and gives the app hint. Once LLBox sets the support phone/WhatsApp/email, the bot reads them automatically.
- DRIVER_LOOKUP_INCLUDE_PHONE. Can be switched off on LLBox's side.
- QA box capacity. Thursday 8, Friday 9 and Saturday 10 October already have 5 deliveries, the box maximum, so registrations on those days are refused (correct behaviour). Our tests use later dates.
- Live dead-box test. The QA box is offline, so a real dead-box request releases the owner's number and notifies them in the app. We tested these flows with stubbed LLBox answers plus one live shape check, and suggest LLBox runs the live COD / dead box / oversized cases on their side.
- Disclosure. On 7 October at 08:29 UTC a bug in our test harness (not in the bot) sent one owner-suite message as an unknown number. The bot correctly treated it as a driver and called driver-contact for NQ4455667788, which LLBox released (request 39). The number stayed inside our test run and was not sent on WhatsApp or stored, but the owner app may have shown a 'driver needs you' notice. The harness is fixed and all suites now stub driver-contact.
- Model fallback. If the classifier times out (6 s) the message is handled by rules only. The common Roman Hindi phrases are covered by rules so they do not depend on the model.
- Greeting numbers. The English greeting uses LLBox's summary text; the Arabic one is built from active_deliveries, so the two can count slightly differently. A localised summary from identify would align them.

Appendix A. Example conversations

Taken from the test runs of 7 October 2026. Six-digit codes are masked. In driver examples the customer number +966511112222 is a stubbed test value, not a real number.

Owner: greeting and menu (F3, F13)

Message	Bot reply
Hi	Hello Khalid This is LLBox. You have 30 deliveries registered, 15 with a driver on the way, a parcel waiting to be collected. I can help you: 1. Register a new delivery 2. Check what's on the way 3. Cancel one you no longer expect 4. Send you a code to open your box 5. Check your box (parcel inside, battery, online) 6. Which number to give couriers 7. Report a problem with your box or a parcel What would you like to do? Reply with a number or just tell me. `outcome: owner_greeting`

Owner: register, COD, status, cancel (F1, F2, F8)

Message	Bot reply
please register X360T78708A	Thanks. Roughly when are you expecting X360T78708A? Today, tomorrow, or a date — or just say you're not sure. `outcome: register_ask_day`
in 9 days	Done X360T78708A is registered for Friday 16 October. When the driver arrives I'll give them a code and let you know. Is it cash on delivery? Reply "COD" (with the amount if you know it), "prepaid", or "not sure". `outcome: registered`
not sure, i'll check	No problem — I've left the payment as unknown. `outcome: payment_unknown`
where is X360T78708A	X360T78708A is registered for Friday 16 October and waiting for the driver. `outcome: reference_status`
cancel X360T78708A	Cancel X360T78708A (expected Friday 16 October)? Reply "yes" to confirm. `outcome: cancel_confirm`
yes	Cancelled X360T78708A won't be accepted at your box. `outcome: cancelled`
tomorrow	Sure — what's the order or tracking number? `outcome: register_ask_ref`

Owner: three references, full day, COD for all (F10, F12)

Message	Bot reply
X360T78708B X360T78708C X360T78708D	Thursday 8 October already has 5 deliveries expected, which is the most your box can take in one day. Would another day work for X360T78708B, X360T78708C, X360T78708D? Or if a parcel is already sitting in the box, collecting it frees up space. `outcome: register_box_capacity_exceeded`
in 10 days	Done X360T78708B is registered for Saturday 17 October. Done X360T78708C is registered for Saturday 17 October. Done X360T78708D is registered for Saturday 17 October. When a driver arrives I'll give them a code and let you know. If any of these is cash on delivery, send its order number with "COD" and the amount. `outcome: registered`
COD 200 riyal for all of them	Noted — X360T78708B is marked cash on delivery (200 SAR). Noted — X360T78708C is marked cash on delivery (200 SAR). Noted — X360T78708D is marked cash on delivery (200 SAR). `outcome: payment_cod`

Owner: dates refused (F10, F11)

Message	Bot reply
register X360T78708E for yesterday	Tuesday 6 October has already passed. When are you expecting X360T78708E? Today, tomorrow, or a date — or just say you're not sure. `outcome: register_day_past`
register X360T78708E in 200 days	Sunday 25 April is more than 120 days away, and I can only register deliveries up to 120 days ahead. When are you expecting X360T78708E? Or just say you're not sure. `outcome: register_day_too_far`

Owner: box, number, codes (F7, F9)

Message	Bot reply
What number do I give Noon?	Give couriers this WhatsApp number: +1 507 710 7553 When a driver messages it with the order number, they get your box location and a one-time code. Use your box's address as the delivery address, and register the order here so the box expects it. `outcome: box_number`
is my box ok? anything inside?	Yes — there's a parcel waiting in your box now. It is about 99% full. 5 deliveries expected today (it comfortably holds 3 a day, 5 at most). The box hasn't checked in since Wednesday 30 September, 15:10, battery 73%, so it may be offline. If it isn't working, tell me and I'll report it. `outcome: box_status`
send me a code for my cleaner	Here's a code to open your box: •••••• Valid for the next hour. It opens the box for whoever you share it with, and doesn't affect any delivery. `outcome: guest_code`
the driver says the code doesn't work for NQ4455667788	Here's a new code for NQ4455667788: •••••• Please give this to the driver. The old code no longer works. Once they enter this one and close the lid, the delivery is complete. `outcome: replacement_code`

Owner: reports, preferences, account (F15, F16)

Message	Bot reply
the box has a red light and won't open	I've reported this, reference T-1. The LLBox team will follow up, and you'll see it in your app. `outcome: issue_reported`
the box has a red light and it will not open at all	I've already reported this a few minutes ago (reference T-1). The team will follow up. `outcome: report_duplicate`
STOP	Done — I won't send you any messages unless you write to me first. Reply START any time to turn them back on. `outcome: preferences_optOut`
I don't want drivers to call me	Done — drivers won't be given your number. If a driver is stuck, I'll let you know in the app instead. `outcome: preferences_driverContactOff`
please delete my account	I can't delete your account from here, so I've passed your request to the LLBox team (reference T-1) and someone will come back to you. `outcome: account_request_passed`
my parcel was stolen from the box	I've reported this, reference T-1. The LLBox team will follow up, and you'll see it in your app. If you believe a parcel was stolen, please also contact the store or courier so they can open a claim. `outcome: issue_reported`

Owner: Arabic

Message	Bot reply
مرحبا	مرحبا Khalid معك LLBox. لديك 48 طلبا نشطا، وطرد بانتظار الاستلام في الصندوق. يمكنني مساعدتك في: 1. تسجيل طلب جديد 2. معرفة الطلبات القادمة 3. إلغاء طلب لم تعد تنتظره 4. إرسال رمز لفتح صندوقك 5. حالة صندوقك (طرد بالداخل، البطارية، الاتصال) 6. الرقم الذي تعطيه لشركات التوصيل 7. الإبلاغ عن مشكلة في الصندوق أو طرد كيف أساعدك؟ أرسل رقم الخيار أو اكتب طلبك. `outcome: owner_greeting`
وش جاي هالأسبوع	هذه الطلبات القادمة إلى صندوقك: • NQ4455667788 — السائق في الطريق الآن • 290055667788 — السائق في الطريق الآن • N77889901A — السائق في الطريق الآن • RACE0410X — السائق في الطريق الآن • X360PH9954 — السائق في الطريق الآن • SPAM0410Q — السائق في الطريق الآن • LBL0410IMG — السائق في الطريق الآن • 7700330022 — السائق في الطريق الآن • BB554400 — متوقع يوم الأحد 11 أكتوبر • AA554400 — متوقع يوم الأحد 11 أكتوبر هل تود إلغاء أي منها؟ `outcome: listed`

Owner with two boxes (F8)

Message	Bot reply
register ABC123456 tomorrow	Which box is this for? 1. Home villa (BOX-11) 2. Office (BOX-12) Reply with the number or the name. `outcome: ask_box`
the office one	Done ABC123456 is registered for Thursday 8 October. When the driver arrives I'll give them a code and let you know. Is it cash on delivery? Reply "COD" (with the amount if you know it), "prepaid", or "not sure". `outcome: registered`
send me a code for my cleaner	Which box is this for? 1. Home villa (BOX-11) 2. Office (BOX-12) Reply with the number or the name. `outcome: ask_box`
office	Here's a code to open your box (Office): •••••• Valid for the next hour. It opens the box for whoever you share it with, and doesn't affect any delivery. `outcome: guest_code`

Driver: Roman Hindi problems (R1)

Message	Bot reply
	The box is responding on our side. Please try the code once more, all six digits, and wait for the beep. If it still won't open, tell me and I'll let the customer know. `outcome: box_online_retry`
	Please don't force the lid — leave the parcel out of the box for SPL77100203 and close the lid. I've let the customer know in their app and they should reply shortly. If you can't wait, please follow your usual process for an undelivered parcel — nothing is recorded against you. `outcome: stuck_notified`
	HANDOVER SPL77100205 I can't share the customer's number for SPL77100205. I've let the customer know in their app and they should reply shortly. If you can't wait, please follow your usual process for an undelivered parcel — nothing is recorded against you. `outcome: stuck_notified`
	Sorry the code isn't working. What's the order number? I'll let the customer know right away. `outcome: stuck_needs_reference`
	The box is responding on our side. Please try the code once more, all six digits, and wait for the beep. If it still won't open, tell me and I'll let the customer know. `outcome: box_online_retry`

Driver: oversized parcel with size (R1, F1)

Message	Bot reply
	Please don't force the lid. What are the parcel's length, width and height in centimetres? For example: 60 x 40 x 30 `outcome: parcel_size_asked`
	Please don't force the lid. Call the customer to arrange the handover: Sara, +966511112222. I've let them know you're calling. Please use this number for this delivery only. `outcome: contact_released`

Driver: cash on delivery (F1, F3)

Message	Bot reply
	This order is cash on delivery (350 SAR). The LLBox can't take cash, so please speak to the customer directly: Sara, +966511112222. I've let them know you're calling. Please use this number for this delivery only. `outcome: contact_released`
	I can help with a cash-on-delivery order. What's the order number? `outcome: cod_needs_reference`
	The customer marked SPL77100212 as already paid, so there's no cash to collect. You can leave it in the box — send me the order number again and I'll give you the box code. `outcome: cod_prepaid`
	The customer prefers not to be called directly. I've told them you need help and they should reply shortly. `outcome: contact_opted_out`
	تسليم SPL77100216 هذا الطلب مسجل كدفع عند الاستلام. إذا كنت تحتاج إلى تحصيل المبلغ أولا، أخبرني وسأوصلك بالعميل. —————— HANDOVER SPL77100216 This order is marked cash on delivery. If you need to collect payment first, tell me and I will connect you with the customer. `outcome: handover`

Driver: dead box (F1)

Message	Bot reply
	Please try the code once more. I'm checking the box and I'll message you here. `outcome: contact_pending_review`
	The box is responding on our side. Please try the code once more, all six digits, and wait for the beep. If it still won't open, tell me and I'll let the customer know. `outcome: box_online_retry`
	Sorry the code isn't working for SPL77100218. I've let the customer know in their app and they should reply shortly. If you can't wait, please follow your usual process for an undelivered parcel — nothing is recorded against you. `outcome: stuck_notified`
	Sorry, the box isn't responding on our side either. Please call the customer to arrange the handover: Sara, +966511112222. I've let them know you're calling. Please use this number for this delivery only. `outcome: contact_released`

Driver: sticker, emoji, '???' (R4)

Message	Bot reply
	لا أستطيع قراءة الملصقات التعبيرية. أرسل لي رقم الطلب من ملصق الطرد، أو صورة للملصق، وسأساعدك. —————— I can't read stickers. Send me the order number from the parcel label, or a photo of the label, and I'll help. `outcome: sticker`
	ممتاز — سأستلمه منك. هل يمكنك إرسال رقم الطلب أو رقم التتبع؟ عادة يكون مطبوعا على ملصق الطرد، أو ظاهرا في تطبيق التوصيل لديك. —————— Great — I'll take it for you. Could you send me the order or tracking number? It's usually printed on the parcel label, or shown in your delivery app. `outcome: ask_reference`
	أرسل لي رقم الطلب من ملصق الطرد (أو صورة للملصق) وسأساعدك. —————— Send me the order number from the parcel label (or a photo of the label) and I'll help. `outcome: fallback`

Appendix B. Every automated check

Original 30 driver scenarios (30 Sep report) (turn-regression.js): 43/43

Check	Message	Outcome	Result
1		opener	PASS
2		already_completed	PASS
3		not_found	PASS
4		completion_claim	PASS
5		ask_reference	PASS
6		already_completed	PASS
7		ask_reference	PASS
8a		already_completed	PASS
8b		already_completed	PASS
9a		cancelled	PASS
9b		cancelled	PASS
9c		expired	PASS
10a		too_short	PASS
10b		not_found	PASS
10c		cancelled	PASS
11		already_completed	PASS
12		already_completed	PASS
13		manipulation_refused	PASS
14		human_request	PASS
15		already_completed	PASS
16		stuck_notified	PASS
17		stuck_needs_reference	PASS
17b		stuck_notified	PASS
17c		stuck_needs_reference	PASS
17d		human_request	PASS
17e		cancelled	PASS
18		owner_claim_refused	PASS
19		stuck_notified	PASS
20		already_completed	PASS
21		already_matched	PASS
22		stuck_needs_reference	PASS
23		stuck_needs_reference	PASS
24		stuck_needs_reference	PASS
25		already_matched	PASS
26		stuck_notified	PASS
27		expired	PASS
28a		fallback	PASS

Check	Message	Outcome	Result
28b		junk_fallback	PASS
28c		fallback	PASS
28d		fallback	PASS
29a		expired	PASS
29b		expired	PASS
30		cancelled	PASS

Driver edge cases: R1, R3, R4, COD, dead box, oversized, privacy (driver-edge-regression.js): 33/33

Check	Message	Outcome	Result
R1a		box_online_retry	PASS
R1b		box_online_retry	PASS
R1c		parcel_size_asked	PASS
R1c2		contact_released	PASS
R1d		stuck_notified	PASS
R1e		stuck_notified	PASS
R1f		stuck_notified	PASS
R1g		contact_released	PASS
R1h		stuck_notified	PASS
R1i		stuck_needs_reference	PASS
R1i2		box_online_retry	PASS
R1j		box_online_retry	PASS
R1k		stuck_notified	PASS
R1l		stuck_notified	PASS
R3a		handover	PASS
R3b		handover	PASS
R3c		handover	PASS
C1		cod_needs_reference	PASS
C1b		cod_prepaid	PASS
C2		contact_opted_out	PASS
C3		contact_not_available	PASS
C4		contact_limited_notified	PASS
C5		handover	PASS
D1		contact_pending_review	PASS
D2		box_online_retry	PASS
D2b		stuck_notified	PASS
D3		contact_released	PASS
O1		parcel_size_asked	PASS
O1b		parcel_fits	PASS
E1		sticker	PASS

Check	Message	Outcome	Result
E2		ask_reference	PASS
E3		fallback	PASS
PRIV		docs with customer phone: 0	PASS

Owner mode F1-F19 on the QA owner account (owner-regression.js): 33/33

Check	Message	Outcome	Result
F3 greeting	Hi	owner_greeting	PASS
F7 menu digit	2	listed	PASS
F1 cancel unknown never registers	cancel ZZZ99887766	cancel_not_possible	PASS
F2 register asks day	please register X360T78708A	register_ask_day	PASS
F2 day answer	in 9 days	registered	PASS
F3-COD not sure	not sure, i'll check	payment_unknown	PASS
F8 status by ref	where is X360T78708A	reference_status	PASS
F1 cancel by ref confirms	cancel X360T78708A	cancel_confirm	PASS
F1 cancel yes	yes	cancelled	PASS
F2 stale ref not reused	tomorrow	register_ask_ref	PASS
F12 three refs	X360T78708B X360T78708C X360T78708D	register_box_capacity_exceeded	PASS
F10 relative day	in 10 days	registered	PASS
F3-COD for all	COD 200 riyal for all of them	payment_cod	PASS
F6 cancel several	cancel X360T78708B, X360T78708C and X360T78708D	cancel_confirm	PASS
F6 cancel no	no	cancel_kept	PASS
F10 past day	register X360T78708E for yesterday	register_day_past	PASS
F11 >120 days	register X360T78708E in 200 days	register_day_too_far	PASS
F9 box number	What number do I give Noon?	box_number	PASS
F13 box status	is my box ok? anything inside?	box_status	PASS
F14 guest code	send me a code for my cleaner	guest_code	PASS
F15 replacement code	the driver says the code doesn't work for NQ4455667788	replacement_code	PASS
F16 report issue	the box has a red light and won't open	issue_reported	PASS
F16 duplicate report	the box has a red light and it will not open at all	report_duplicate	PASS
F17 STOP	STOP	preferences_optOut	PASS
F17 START	START	preferences_optIn	PASS
F17 no driver calls	I don't want drivers to call me	preferences_driverContactOff	PASS
F18 account change	please delete my account	account_request_passed	PASS

Check	Message	Outcome	Result
F19 theft	my parcel was stolen from the box	issue_reported	**PASS**
F4 self driver escape	I am the driver delivering to this box	ask_reference	**PASS**
F5 wat can you do	wat can you do	owner_menu	**PASS**
F3 Arabic greeting	مرحبا	owner_greeting	**PASS**
F7 Arabic list	وش جاي هالأسبوع	listed	**PASS**
thanks	thanks!	thanks	**PASS**

Two-box owner (F8) and R4 wording (multibox-regression.js): 12/12

Check	Message	Outcome	Result
F8 register asks box	register ABC123456 tomorrow	ask_box	**PASS**
F8 box by name	the office one	registered	**PASS**
F8 box by number	register XYZ998877 for Sunday	ask_box	**PASS**
F8 box by number answer	1	registered	**PASS**
F8 box named up front	register QWE445566 at home tomorrow	registered	**PASS**
F8 guest code asks box	send me a code for my cleaner	ask_box	**PASS**
F8 guest code office	office	guest_code	**PASS**
F8 report asks box	the box has a red light and won't open	ask_box	**PASS**
F8 report home	1	issue_reported	**PASS**
R4 Urdu script	میرا پاس پارسل ہے، باکس کہاں ہے؟	ask_reference	**PASS**
R4 spaced form same driver	spl 88 123456	handover	**PASS**
R4 same ref compact	SPL88123456	handover	**PASS**

Appendix C. Files changed

File	Role
backend/services/turn.service.js	Pipeline: dedupe, read, identify, route, driver flow, reply
backend/services/owner.flow.js	Owner mode: menu, intents, slots, register, list, cancel, codes, box, reports, preferences
backend/services/contact.flow.js	driver-contact: COD, dead box, oversized parcel, conversation_id
backend/services/followup.service.js	Background worker: delivery-status thank-you and contact-review polling
backend/services/issue-classifier.service.js	Model classifier for driver problems (classification only)
backend/services/inbound.service.js	References (spaced join), language, problem and owner intent rules, dates, sizes
backend/services/day-format.js	Day and time wording in English and Arabic (Riyadh time)
backend/services/templates.service.js	All reply wording, English and Arabic
backend/services/llbox.service.js	LLBox API client incl. all Part F endpoints
backend/models/TurnState.js, FollowUpJob.js	Conversation memory; follow-up jobs (never hold a phone number or code)
n8n/fix_operations_workflow.py	Workflow patch: sequential photo sends, reaction skip
backend/scripts/*-regression.js	Four regression suites (121 checks)